

Toward a Total Causal Error Framework: Extending Cornfield's Insights Across Causal Identification, Representation, and Measurement

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Abstract

Cornfield's 1959 analysis of smoking and lung cancer showed how explicit assumptions about unobserved confounding and selection mechanisms shape the boundary between statistical association and causal conclusions. Today, causal inference spans diverse fields including economics, statistics, psychology, epidemiology, and evaluation, with distinct traditions and tools. Without a shared framework, it can be difficult to accumulate evidence across studies or to diagnose why results differ when similar questions are studied independently, particularly when differences arise from upstream design and data-generation choices rather than analytic methods alone. While tools for causal identification such as Directed Acyclic Graphs (DAGs) have become widely used across disciplines, they do not naturally foreground threats arising from representation and measurement errors. In contrast, the Total Survey Error (TSE) framework enumerates representation and measurement errors across the survey life cycle and is routinely used to evaluate tradeoffs in survey design and data collection. We propose a Total Causal Error (TCE) framework that integrates causal inference with the TSE perspective to organize risks to causal conclusions along parallel paths of causal identification, representation, and measurement, across the stages of the theoretical estimand, empirical estimand, data collection, data processing, and estimation and interpretation. We propose a version of this framework, provide explicit definitions for its components, illustrate a small number of use cases, and invite further development and coordination across fields.

Keywords: causal inference, total survey error, causal estimand, study design, causal identification, sensitivity analysis, directed acyclic graph, selection bias, design tradeoffs, measurement error

1. Introduction

Cornfield’s report on smoking and lung cancer synthesized evidence across multiple study designs to directly engage major critiques about selection, measurement, and omitted variables (Cornfield et al., 1959). While this work helped establish standards such as relative risk and sensitivity analysis (Greenhouse, 1982), it was largely organized as a response to criticisms rather than as an explicit framework for anticipating and diagnosing threats to causal inference. In many applied settings, causal claims do not receive the same level of scrutiny; researchers must therefore anticipate vulnerabilities and address them through design and analysis decisions.

Given the diversity of disciplines using causal inference, a unifying framework must be accessible enough to be broadly understood, yet rich enough to support distinct research traditions. Star and Griesemer (1989) described such coordinating tools as boundary objects. Directed Acyclic Graphs (DAGs) provide a prominent example: they are widely used across fields and serve as a powerful planning tool for causal identification (Hernán and Robins, 2020). However, DAGs are primarily designed to reason about causal structure and do not naturally represent errors related to representation (e.g., limited common support) or measurement (e.g., construct validity).

An enduring boundary object for these latter concerns is the Total Survey Error (TSE) framework, which systematizes the sources of measurement and representation error across the survey life cycle (Groves and Lyberg, 2010). TSE has been extended beyond traditional surveys to multisource and administrative data settings (Rocci et al., 2022; Biemer and Amaya, 2020; Reid et al., 2017), and has motivated broader “Total Error” perspectives (Lavrakas, 2013). Nevertheless, most Total Error frameworks remain oriented toward descriptive population quantities (e.g., population means) rather than causal estimands, creating a mismatch as surveys and other observational data are increasingly used for causal inference.

On the other hand, survey research and causal inference have developed closely related methodologies. Calibration and doubly robust methods appear across survey sampling and causal inference (Bang and Robins, 2005; Zhang et al., 2023), while causal frameworks have long been used to model selection and nonresponse to surveys (Fay, 1986; Kenah, 2023). This convergence highlights the need for a framework that explicitly integrates causal inference and survey research traditions.

In this commentary, we propose a Total Causal Error (TCE) framework that integrates causal identification with representation and measurement concerns. In their synthesis of retrospective and prospective studies, Cornfield et al. organized critiques around selection of study groups, accuracy of information, and multiple variables. These categories align closely with the representation, measurement, and causal identification paths in TCE. By situating estimands and data quality within a unified structure, TCE provides a shared language for collaborative reasoning about where assumptions enter the study process, how errors arise, and what tradeoffs shape causal conclusions.

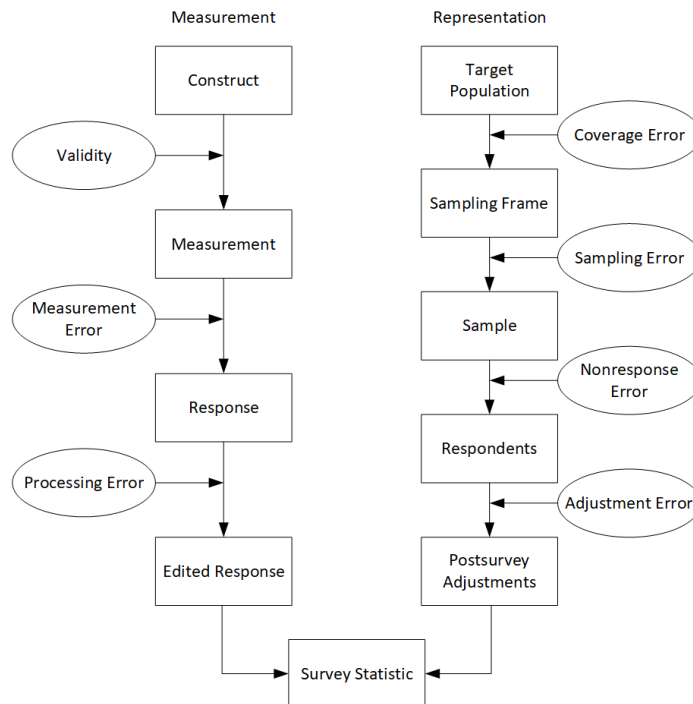


Figure 1: Total Survey Error framework, reproduced from Groves et al. (2011).

2. Reviewing the Total Survey Error Framework

We briefly revisit the Total Survey Error (TSE) framework to clarify what it contributes, and what it omits, for causal inference. Developed in response to early survey theory’s narrow focus on sampling error (Deming, 1944), TSE provides a structured way to conceptualize how representation and measurement errors can cause survey estimates to differ from target population quantities (Groves and Lyberg, 2010; Lyberg and Stukel, 2017). The TSE framework, depicted in Figure 1, enumerates multiple sources of error across the survey life cycle and discourages exclusive focus on any single component, such as sampling or non-response, in isolation. A well-known example of the error tradeoffs made explicit by TSE is motivated underreporting in screening interviews, where efforts to reduce measurement error can increase nonresponse error, and vice versa (Tourangeau et al., 2012).

TSE can be used both prospectively, for comparing design and analytic options before results are finalized, and retrospectively, for interpreting estimates in light of the different sources of error that may have entered across the analytic process. The design-centered perspective of TSE is directly relevant to causal inference, as many threats to causal validity arising from selection, confounding, measurement, and missing data are determined before analysis begins through how data are generated and assembled (Rubin, 2008).

TSE provides a structured vocabulary for evaluating how representation and measurement quality are shaped by upstream design choices, but it is primarily oriented toward descriptive estimation. In causal analyses, similar design decisions often implicitly redefine the causal question itself, including who the results apply to, what variation identifies the

effect, and how observed measures relate to the underlying causal phenomenon, such that downstream analytic rigor cannot recover the intended causal meaning. This motivates the need for a framework that retains TSE’s strengths while explicitly incorporating causal identification and estimand-based reasoning.

3. Introducing the Total Causal Error Framework

We propose an interpretable, design-first framework that provides a structure within which we can reason about threats to causal conclusions. Extending ideas from TSE to causal inference, the TCE framework, depicted in Figure 2, organizes potential sources of error in causal conclusions along three parallel paths:

- **Representation:** Who is included in the study and how inclusion is adjusted to approximate a target population.
- **Causal identification:** What variation and assumptions support a causal contrast.
- **Measurement:** How well constructs are operationalized, recorded, and processed.

Across these paths, TCE classifies sources of error by the stage of the study life cycle at which they arise:

- **Theoretical estimand:** A precise specification of the causal effect of interest, defined by a target population and counterfactual contrast.
- **Empirical estimand:** The causal quantity that is identifiable from observed data for a specific sample and identification strategy, under stated assumptions.
- **Data collection:** How units participate in the study, experience exposures, and are measured in practice, translating design choices into observed data.
- **Data processing:** Post-collection transformations and adjustments that encode analytic assumptions and modify the data prior to estimation.
- **Estimation and interpretation:** Quantifying causal effects and uncertainty from processed data and assessing the causal claims supported by the study design and assumptions.

Boxes in the diagram represent TCE components, corresponding to potential sources of error in causal conclusions. Arrows between components indicate the typical direction in which design decisions propagate constraints through the study process. They do not imply a strictly one-way sequence; earlier components may need to be revisited in response to later feasibility or validity concerns.

The TCE framework is intended not as a checklist of errors to be minimized independently, but as a diagnostic tool for reasoning about how design choices made at different stages jointly determine the causal claims the data can support. Figure 2 can be read as a map for orienting design decisions across the study life cycle: moving horizontally traces how upstream design choices constrain downstream components, while moving vertically

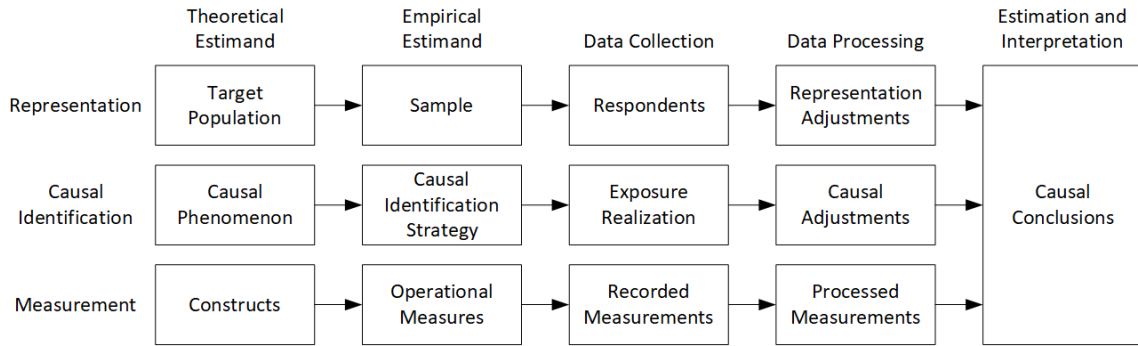


Figure 2: Total Causal Error (TCE) framework.

highlights design considerations across representation, causal identification, and measurement.

Rather than exhaustively cataloging each component, we illustrate how the framework can be used through three vignettes drawn from Cornfield et al. (1959). These examples were selected to highlight recurring tensions in applied causal research and to assess the consequences of departures from idealized alignment between components. Table 1 defines each TCE component and clarifies its role within the framework.

3.1 Vignette 1: Narrowing the target population to strengthen internal validity

In practice, debates about external validity or policy relevance often arise along the representation path, particularly when there are questions about how well the observed data represent the population to which the analysis seeks to generalize. By contrast, threats to internal validity often occur along the causal identification and measurement paths. In many studies, design choices that weaken one dimension can strengthen another.

One example occurs when features of the data or design threaten credible causal comparisons for the full population of interest. In those cases, researchers may redefine the target population to a narrower subpopulation for which the comparison is more defensible. This can strengthen internal validity, though the resulting estimate pertains to a narrower population than the one initially motivating the study. When causal phenomena are in question, however, narrowing the target population can help rule out competing counterfactual hypotheses.

Cornfield (1959) highlighted this point in discussing the Doll and Hill (1956) study, in which the target population was restricted to British physicians. This design choice narrowed the scope of inference but reduced heterogeneity related to socioeconomic status and occupation, enabled near-complete outcome ascertainment, and improved the quality of outcome measurement, as clinical and pathological evidence used to classify cause of death was presumably more reliable than in broader population studies. The resulting estimate applied to a more limited population, but one for which the causal claim was more clearly defensible.

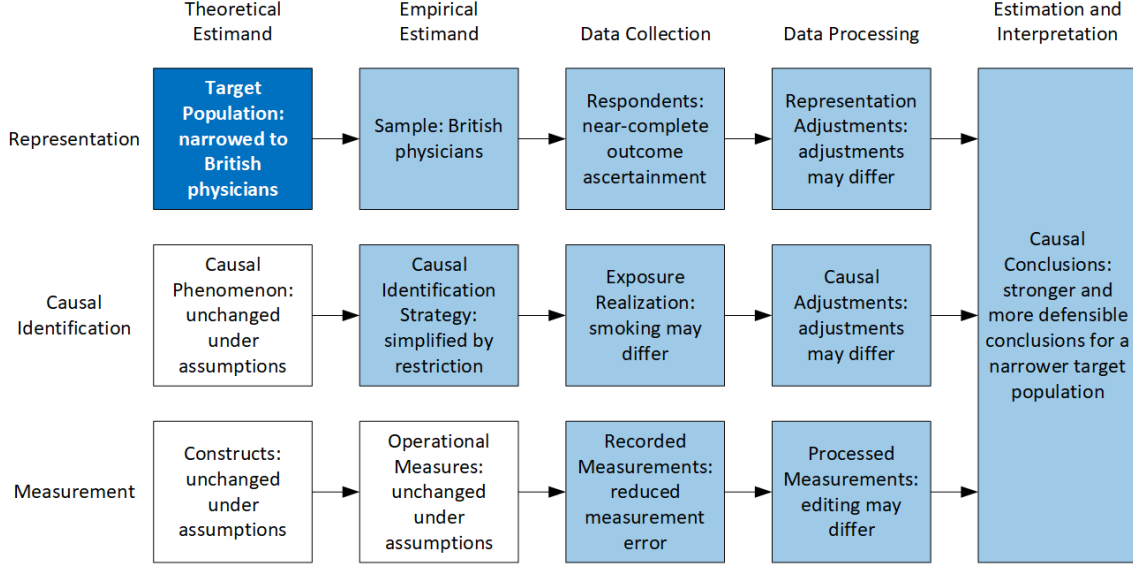


Figure 3: The TCE showing changes resulting from the design decision proposed in Vignette 1. Dark blue shading indicates components that change directly, light blue shading indicates components that are affected indirectly, and unshaded components remain unchanged.

More broadly, this restricted estimand provided vital evidence for the larger question of smoking and lung cancer that concerned Cornfield. Although insufficient to resolve the question on its own, the study constrained the space of plausible alternative explanations, strengthening the cumulative argument across studies with different target populations.

3.2 Vignette 2: Using design to address selection on exposure

Although exposure is specified in the empirical estimand, the data collection stage determines how that exposure is realized in practice. Cornfield, citing a critique by Neyman, explained how cross-sectional studies of prevalent lung cancer cases could introduce bias through selection on exposure induced by survival. In such designs, if smokers with lung cancer outlived nonsmokers with lung cancer, exposed individuals could be overrepresented among observed cases simply because they remained observable for longer. The DAG in Figure 4 illustrates this argument: in a cross-sectional study of prevalent cases, inclusion implicitly conditions on survival (S), opening a noncausal path between the exposure (X) and outcome (Y).

Rather than treating this as a flaw to be corrected analytically, Cornfield pointed to a study design that eliminated this source of error. Prospective designs implement follow-up consistently across units, eliminating the selection and exposure-realization problems inherent in cross-sectional studies. Figure 5 shows how this design choice relates to TCE components, affecting exposure realization and sample selection directly, and some other components indirectly.

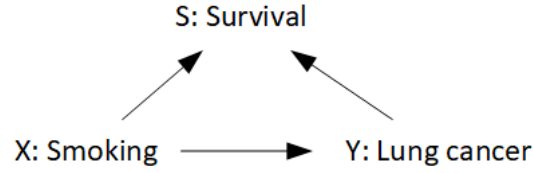


Figure 4: Causal DAG for cross-sectional studies of prevalent cases, where inclusion implicitly conditions on survival (S), a collider of smoking (X) and lung cancer (Y).

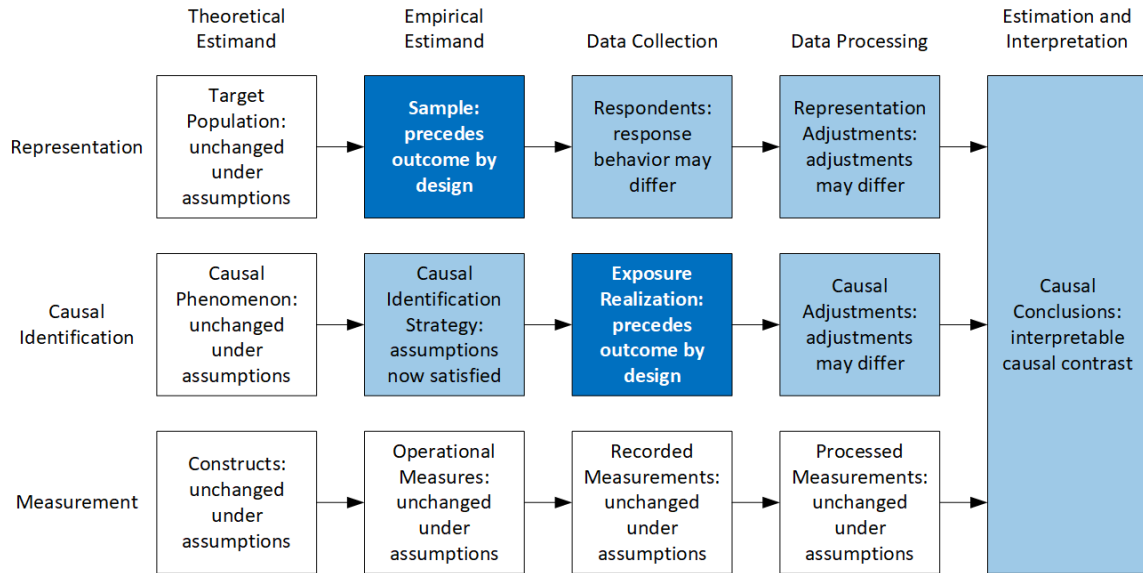


Figure 5: The TCE showing changes resulting from the design decision proposed in Vignette 2. Dark blue shading indicates components that change directly, light blue shading indicates components that are affected indirectly, and unshaded components remain unchanged.

3.3 Vignette 3: Reasoning along the measurement path

The TCE framework represents measurement as a path from constructs to operational measures, recorded measurements, and processed measurements. Rather than treating measurement error as a single source of bias, Cornfield assessed the amount of error in recorded measurements, its likely consequences for conclusions, and the alignment between final processed measurements and the construct.

In the early debates over smoking and lung cancer, many arguments centered on possible measurement errors. Under the heading “Accuracy of Information,” Cornfield emphasized that both disease diagnosis and self-reported smoking behavior were subject to error, but noted that the severity and implications of this error differed by study design. Diagnostic accuracy was less controllable in prospective studies, where all deaths had to be classified, than in retrospective studies, where case confirmation was possible. Errors in reporting smoking behavior were also acknowledged, but evidence suggested that most misclassification was minor.

Cornfield reasoned about how different patterns of error would affect observed associations. Misclassification of cause of death would not simply weaken or strengthen results uniformly: if lung cancer deaths among smokers were attributed to other causes, the smoking–lung cancer association would be overstated while associations with other diseases would be understated, and the reverse would hold if deaths due to lung cancer were misclassified as other causes. In this way, different realizations of error would redistribute risk across outcomes rather than arbitrarily distort the causal conclusions.

Measurement error concerns how well values are recorded; construct validity concerns whether those values correspond to the underlying construct of interest. In addition to assessing the magnitude and direction of measurement error, Cornfield examined whether the available measures adequately preserved the relevant exposure and outcome constructs. He drew on evidence that smoking questionnaires aligned with tobacco production and taxation data and that the smoking–lung cancer association was stronger for epidermoid carcinoma than for adenocarcinoma, a pattern he argued could not plausibly be generated by reporting or interviewer bias and that supported the adequacy of the operational outcome construct. When construct validity is preserved, the TCE framework provides a starting point to trace how imperfect measurement relates to different components of the study and assess whether it meaningfully alters the causal conclusion.

4. Discussion

We begin by summarizing the central contributions of the Total Causal Error (TCE) framework. Motivated by the complementarity of survey research and causal inference traditions, TCE provides a shared language for interdisciplinary collaboration, connects causal identification to data quality considerations, and underscores a central insight: causal conclusions are more fatally threatened by upstream design choices than downstream analytic adjustments. Survey research benefits from more explicit causal reasoning, while causal analyses benefit from deeper engagement with survey-based sources of error.

A second contribution of TCE is greater precision in how sources of error are described. Terms such as “selection bias” or “measurement error” are often used generically, even though distinct mechanisms arise under these labels at different stages of the study process

and have different implications for causal interpretation. TCE distinguishes, for example, between selection into the sample, which is primarily a representation issue; selection on exposure, which may arise during data collection stage to threaten causal identification, and explicitly conditioning on post-treatment variables, a failure in causal adjustment. Making these distinctions helps clarify which assumptions are being invoked and where corrective tools operate in the study life cycle.

We next highlight several directions in which TCE suggests productive extensions of current practice. Integrated with common causal tools such as outcome modeling and propensity score methods, sampling and weighting approaches from the TSE tradition offer a way to strengthen both internal and external validity in observational studies. Cornfield’s discussion of Doll and Hill’s study illustrates this integration. Their use of a probability sample from the population served by the local hospital supported the argument that smoking habits were adequately represented among the patients recruited into the study, reinforcing the credibility of the causal comparison.

A second avenue for future work concerns sensitivity analysis as a unifying tool across survey and causal inference traditions. The Cornfield conditions evaluate how strong an unmeasured confounder would need to be to account for an observed association, while TSE-based sensitivity analyses assess how alternative assumptions about selection processes affecting representation alter resulting inferences. Within the TCE framework, these approaches can be combined by systematically varying assumptions across error sources, rather than relying on a single set of weights or a single ignorability assumption, to assess the robustness of treatment effect estimates to plausible departures from idealized conditions.

We conclude by noting several caveats and considerations that shape the interpretation and use of TCE. Although TSE labels a standard set of errors for each component, TCE does not impose a one-to-one taxonomy linking error mechanisms to individual components. We have organized threats by where they arise in the study process and by their dominant mechanisms, but the same process may affect multiple paths simultaneously. For example, selection processes may alter both representation and causal identification, and measurement limitations may interact with selection mechanisms in ways that cannot be cleanly decomposed.

Finally, it is important to emphasize what “total” in TCE does and does not imply. “Total” does not mean that all errors can be enumerated, quantified, or independently corrected. Design choices that reduce one source of error may exacerbate another. A pragmatic implication of TCE is therefore not to promise comprehensive error quantification, but to guide researchers toward identifying the most consequential error sources in a given context, exploring their interactions, and being explicit about what remains unresolved. In this sense, TCE extends Cornfield’s insight that credible causal claims depend on confronting alternative explanations, by offering a systematic way to do so across modern causal workflows.

References

Heejung Bang and James M Robins. Doubly robust estimation in missing data and causal inference models. *Biometrics*, 61(4):962–973, 2005. doi: 10.1111/j.1541-0420.2005.00377.x.

- Paul P Biemer and Ashley Amaya. Total error frameworks for found data. In *Big Data meets survey science: A collection of innovative methods*, pages 131–161. Wiley, 2020.
- Jerome Cornfield, William Haenszel, E Cuyler Hammond, Abraham M Lilienfeld, Michael B Shimkin, and Ernst L Wynder. Smoking and lung cancer: Recent evidence and a discussion of some questions. *Journal of the National Cancer Institute*, 22(1):173–203, 1959.
- W Edwards Deming. On errors in surveys. *American Sociological Review*, 9(4):359–369, 1944.
- Richard Doll and Austin Bradford Hill. A study of the aetiology of carcinoma of the lung. *British Medical Journal*, 2:1271–1286, 1952.
- Richard Doll and Austin Bradford Hill. Lung cancer and other causes of death in relation to smoking: A second report on the mortality of British doctors. *British Medical Journal*, 2:1071–1081, 1956.
- Robert E Fay. Causal models for patterns of nonresponse. *Journal of the American Statistical Association*, 81(394):354–365, 1986.
- Samuel W Greenhouse. Jerome Cornfield’s contributions to epidemiology. *Biometrics*, pages 33–45, 1982.
- Robert M Groves and Lars Lyberg. Total survey error: Past, present, and future. *Public Opinion Quarterly*, 74(5):849–879, 2010.
- Robert M Groves, Floyd J Fowler Jr, Mick P Couper, James M Lepkowski, Eleanor Singer, and Roger Tourangeau. *Survey methodology*. John Wiley & Sons, 2011.
- Miguel A Hernán and James M Robins. *Causal Inference: What If*. Chapman & Hall/CRC, Boca Raton, 2020.
- Eben Kenah. A potential outcomes approach to selection bias. *Epidemiology*, 34(6):865–872, 2023. doi: 10.1097/EDE.0000000000001660.
- Paul J Lavrakas. Presidential address: Applying a total error perspective for improving research quality in the social, behavioral, and marketing sciences. *Public Opinion Quarterly*, 77(4):831–850, 2013.
- Lars E Lyberg and Diana Maria Stukel. The roots and evolution of the total survey error concept. In *Total survey error in practice*, pages 1–22. Wiley, 2017.
- Giles Reid, Felipa Zabala, and Anders Holmberg. Extending TSE to administrative data: A quality framework and case studies from Stats NZ. *Journal of Official Statistics*, 33(2):477–511, 2017.
- Fabiana Rocci, Roberta Varriale, and Orietta Luzi. Total process error: An approach for assessing and monitoring the quality of multisource processes. *Journal of Official Statistics*, 38(2):533–556, 2022.

- Donald B Rubin. For objective causal inference, design trumps analysis. *Annals of Applied Statistics*, 2(3):808–840, 2008. doi: 10.1214/08-AOAS187.
- Susan Leigh Star and James R Griesemer. Institutional ecology, ‘translations’ and boundary objects: Amateurs and professionals in Berkeley’s Museum of Vertebrate Zoology, 1907–39. *Social Studies of Science*, 19(3):387–420, 1989.
- Roger Tourangeau, Frauke Kreuter, and Stephanie Eckman. Motivated underreporting in screening interviews. *Public Opinion Quarterly*, 76(3):453–469, 2012.
- Shixiao Zhang, Peisong Han, and Changbao Wu. Calibration techniques encompassing survey sampling, missing data analysis and causal inference. *International Statistical Review*, 91(2):165–192, 2023.

Table 1: Components of the Total Causal Error (TCE) framework across stages of a causal study.

Stage	Component	Definition (what is specified)	Key design question	Dominant error concerns
Theoretical Estimand	Target Population	Population to which the causal claim is intended to apply	To whom does the effect pertain?	Ambiguous scope of inference, limited external validity
	Causal Phenomenon	Underlying causal process generating outcome differences	What causal process is being posited?	Misspecified mechanisms
	Constructs	Conceptual variables representing the causal process (e.g., exposures, outcomes, confounders, mediators)	What concepts must be represented?	Ambiguity in the definition of causal constructs
Empirical Estimand	Sample	Set of observed units over which the empirical estimand is defined	Which units enter the analysis?	Selection into the sample
	Causal Identification Strategy	Assumptions under which observed variation identifies causal effects	When is observed variation causal?	Confounding, interference, misspecification of causal pathways
	Operational Measures	Observable variables used to represent constructs	How are concepts operationalized?	Construct validity or proxy validity
Data Collection	Respondents	Sampled units with observed data	Who provides information?	Nonresponse bias
	Exposure Realization	Assignment, receipt, timing, and dose of the intervention experienced by units	How is treatment received?	Selection on exposure, immortal time bias
	Recorded Measurements	Observed values of operational measures	What is observed?	Measurement and reporting error
Data Processing	Representation Adjustments	Weighting or restriction to address lack of overlap or representativeness	How is representativeness or overlap addressed?	Weight instability, unintended redefinition of the target population
	Causal Adjustments	Conditioning, weighting, matching, or model-based adjustment to address confounding	How is confounding addressed?	Residual confounding, conditioning on post-treatment variables (e.g., colliders)
	Processed Measurements	Edited or transformed versions of recorded measurements	How are measurements altered?	Coding or transformation error
Estimation & Interpretation	Causal Conclusions	Claims drawn from estimated effects and uncertainty under stated assumptions	What causal claims are justified by the evidence?	Overinterpretation, misalignment with the estimand